High-Resolution SAR Image Classification via Multiscale Local Fisher Patterns

Zhiqiang Zhao, Meng Jia, and Lei Wang

***Abstract*— With the increased number of synthetic aperture radar (SAR) systems that have been launched in recent years, many high-quality images have been produced for precise earth observation. The number of available images requires new feature extraction techniques to capture enough discriminant information from high-resolution SAR images. To effectively characterize different land covers, a multiscale local Fisher pattern (MLFP) is proposed to extract both statistical and spatial properties in this work. First, the Fisher vector (FV) of each pixel is calculated after the statistical distribution of high-resolution SAR intensities is modeled via a Gamma mixture model. Then, the generative and discriminant information implied in the FV is exploited after the extraction of the geometrical information between neighboring pixels within different scales. Two experiments are conducted to demonstrate the performance of the proposed MLFP, which are high-resolution SAR image land-cover classification and SAR scene classification. Compared with some existing approaches, the proposed MLFP shows superior performance after the statistical and spatial structure information from different land covers is exploited.**

***Index Terms*— Fisher vector (FV), Gamma mixture model, high-resolution synthetic aperture radar (SAR) image, land-cover classification, multiscale local Fisher pattern (MLFP).**

I. Introduction

AS ONE kind of active sensor with timeless, high-resolution, wide-coverage, and all-weather imaging capability, synthetic aperture radar (SAR) systems have been launched in many variations in recent years [1]–[3]. SAR systems can provide rich images of exact earth observation in various military and civilian applications, such as disaster monitoring, crop yield mapping, and automatic target detection [4]–[7]. This has attracted great interest from many organizations and countries to develop multiple airborne and spaceborne platforms to acquire SAR images with different spatial resolutions. With an increase in resolution, it is more common to find that a single-resolution cell, compared with images with medium or low resolution, may contain only a few scatters or even just one dominant scatter. Because various objects can be observed clearly by modern SAR sensors, a solid comprehension of SAR images requires that different land covers are clearly defined and identified [8], [9]. For most existing SAR image classification systems, the performance relies heavily on the significant feature descriptors and high-quality labeling samples to train a robust classifier. However, compared with classifier model training, learning efficacious representations to exploit discriminant information of different land covers plays a critical role in SAR image classification tasks.

After decades of extensive and in-depth research, several descriptors have been proposed to effectively characterize the geometric, radiometric, and structural features of SAR images. Texture descriptors [10], [11] are widely utilized to describe the statistical characteristic of intensity values of different land covers. Texture descriptors include higher order statistics generated from the gray-level cooccurrence matrix (GLCM), such as the mean, homogeneity, contrast, energy, and entropy. Although they have an excellent capacity to characterize the global properties of different SAR land covers, the complicated structures within a local region are becoming more critical for SAR image processing at increased resolution. Because they can provide valuable edge features of different textures, the Gabor filter bank [12] and wavelet transform [13], [14] with some predefined scales and orientations are employed to extract multiscale and multiorientation features. However, it is confusing to determine a suitable number of scales and orientations of these filters and transforms to capture enough discriminant information to identify different land covers. Dumitru and Datcu [15] studied the dependence of information extraction technique performance on SAR imaging parameters and some selected features. Their research indicated that a heavy dependence exists between the land-cover classification accuracy and various imaging parameters. In addition, these features, which are also called low-level features, fail to characterize SAR scenes because they are incapable of incorporating some prior knowledge to capture enough abstract representations to distinguish different land covers.

Unlike low-level feature descriptors, many mid- and high-level features have emerged to learn abstract representations with the development of machine learning techniques, especially deep learning. After the sparse prior

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on low-level feature is integrated, it can be noticed that mid-level features learned from sparse representation have a better effect than bag-of-models [16], [17] in the characterization of different land covers. Zhou *et al.* [18] proposed a multilevel reconstruction-based multitask joint sparse representation approach to overcome the limits of template-matching-based target recognition with strong background clutter and a noisy environment. Meanwhile, Gu *et al.* [19] proposed a multifeature weight sparse graph to combine multiple types of features for SAR image classification. The results of these works show that sparse representation has excellent advantages of noise robustness, data adaptiveness, and classification performance. Nevertheless, the optimization problem of sparse representation is nonconvex and needs a large amount of runtime to obtain an approximate solution. Furthermore, to improve the representation capability of features, various deep learning techniques have been successfully applied in SAR image processing to automatically learn features for different land covers [20]–[22]. Especially, after combining with ensemble learning and a deep belief network (DBN) in an unsupervised manner, Zhao *et al.* [23] proposed a discriminant DBN model to distinguish different land covers. Liu *et al.* [24] proposed a stacked Fisher autoencoder network to alleviate the influence of speckle in the detection of the changing parts in SAR images. In [25], a statistical convolutional neural network (CNN) was applied for SAR image land-cover classification after the incorporation of the statistical information into a classic CNN framework. In addition to these works, other deep frameworks, such as the deep cascade network [26], [27], deep recurrent encoding network [28], [29], and deep transfer learning [30], [31], have been successfully utilized in the field of SAR image processing to extract features for different tasks. Although these frameworks can effectively extract abstract and high-level features for SAR image processing, a large amount of runtime is required for deep network training, and many high-quality training samples are required to build a robust model of different land covers. In addition, the statistical distribution of intensities should be introduced to model the generative characters of backscatters of different land covers.

To precisely describe the generative process of an SAR image, various probability density functions are employed to extract statistical features of backscatters from different land covers, such as Weibull, Log-normal, Rayleigh, Gamma, and $\mathcal{G}_0$ [32]–[34]. Compared with images with a medium or low resolution, a reduced number of scatters can be observed in a single-resolution cell in images with a higher resolution. Therefore, a simple statistical model shows clear limitations for complicated structures that appear in high-resolution SAR images. In recent years, several methods have been introduced that combine basic probability distributions to build a stronger model to characterize the complex-valued statistical properties of high-resolution SAR images. As the simplest method, many probabilistic mixture models have been introduced and widely used to characterize different land covers, including the Gaussian mixture model [35], generalized Gamma mixture model [36], Wishart mixture model [37], and $\mathcal{G}_p^0$ mixture model [38]. With these mixture models, the complicated spatial structures of high-resolution SAR images can be characterized effectively by combining different subcomponents, which are optimized via an expectation–maximization (EM) procedure. Nevertheless, determining a suitable number of subcomponents for different land covers is also a potentially confusing task. In addition, after the SAR image in the complex-valued wavelet domain was described, a weighted sparse-representation-based classification method [39] was proposed for automatic target recognition after the modeling of the complex-valued subbands with the Weibull or Gamma distributions. Meanwhile, Liu *et al.* [40] proposed a discriminant statistical dictionary learning approach for SAR target configuration recognition by minimizing differences within the same target configuration and maximizing differences among different configurations. The combinations of the discriminant capacity of sparse representation and statistical models provide competitive performance in characterizing different land covers. However, the optimization process of compact dictionary learning is also a time-consuming task.

With a learned probability model, a Fisher kernel [41], [42] can be generated by combining the strengths of generative and discriminant approaches, which is widely used in the field of computer vision for characterizing different contents of images. Especially, a gradient vector is derived from the probability density function that models the statistical characterization of the considered image data. By integrating a Potts-like energy model, Redolfi *et al.* [43] derived a Fisher vector (FV) from the real Wishart mixture densities for pixelwise polarimetric SAR image classification. This allowed that the proposed FV can obtain encouraging results on two challenging data sets by considering the spatial dependencies between neighboring regions in polarimetric SAR images. Furthermore, Lin *et al.* [44] proposed a superpixel-level FV based on a Gaussian mixture model for high-resolution SAR image ship detection by characterizing the difference between superpixels from ship and clutter areas. Compared with other models, the FV can achieve an effective and robust detection performance under low signal-to-clutter conditions. Although the FV can effectively capture enough statistical information to distinguish different land covers from SAR images, an appropriate and accurate statistical model for SAR intensities is a critical factor. In addition, some geometrical constraints should be considered to improve the representation capability of the FVs.

As one of the classic methods of incorporating the local geometrical information, the local binary pattern (LBP) vector has been widely used in the field of computer vision to measure the local contrast for image texture analysis, microtexton searches in a local region, and so on [45]–[47]. Inspired by LBP, a multiscale local Fisher pattern (MLFP) is proposed in this work to characterize different land covers of high-resolution SAR images after the spatial dependencies between the neighboring pixels are taken into consideration. The major contributions of this article are as follows.

1) Though the modeling of the probability distribution of high-resolution SAR intensities via a Gamma mixture model (ΓMM), a FV is derived for each pixel to fully exploit its generative and discriminant properties.

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2) With the pixelwise FVs, an MLFP is generated by introducing the geometrical information from its neighboring pixels.

To evaluate the performance of the proposed MLFP, two typical high-resolution SAR image processing problems are addressed in this work: high-resolution SAR image land-cover classification and SAR scene labeling. Compared with other approaches, the experimental results demonstrate that the proposed MLFP has excellent advantages in distinguishing different land covers from high-resolution SAR images.

The remainder of this work is organized as follows. Section II introduces the computation procedure of the MLFP based on a Gamma mixture model. Then, the data and experimental results of the proposed approach are presented in Section III. Finally, Section IV concludes this work.

II. Multiscale Local Fisher Pattern

In this section, we introduce the MLFP for high-resolution SAR image classification. Especially, the ΓMM is introduced to model the statistical distribution of intensity values in the first part. Then, a FV is generated to characterize different land covers by encoding the generative and discriminant properties of each pixel. After the incorporation of the geometrical information between neighboring pixels, an MLFP is computed to characterize different land covers in the third part. Finally, some discussions are addressed in the last part of this section.

A. Gamma Mixture Model for High-Resolution SAR Images

Given a set of N high-resolution SAR image pixels, $I = (x_1, x_2, \ldots, x_N)$, the probability density function can be modeled via a Gamma mixture model with K-subcomponents as follows [48], [49]:

$$f(x; \boldsymbol{\psi}) = \sum_{k=1}^{K} \lambda_k g(x; \boldsymbol{\theta}_k) \tag{1}$$

where x_n represents intensity value of the nth pixel. $\boldsymbol{\psi} = \{\lambda_1, \ldots, \lambda_K, \boldsymbol{\theta}_1, \ldots, \boldsymbol{\theta}_K\}$ represents the model parameter. λ_k's are mixing properties of each subcomponent, which should meet $\lambda_k \geq 0$ and $\sum_{k=1}^{K} \lambda_k = 1$ for probability synthesis. Each subcomponent $g_k(x; \boldsymbol{\theta}_k)$ is a Gamma probability distribution with shape parameter $\alpha_k > 0$ and scale parameter $\beta_k > 0$, which is given by

$$g(x; \boldsymbol{\theta}_k) = \frac{x^{\alpha_k - 1} e^{-x/\beta_k}}{\beta_k^{\alpha_k} \Gamma(\alpha_k)} \text{ for } x > 0$$

where $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} \mathrm{d}t$ is the complete Gamma function and $\boldsymbol{\theta}_k = \{\alpha_k, \beta_k\}$ are the parameters of this subcomponent.

To obtain a precise estimation of the parameter $\boldsymbol{\psi}$, the update equation can be derived by a maximum likelihood approach. Especially, the likelihood function of the given intensity set I can be written as

$$\mathcal{L}(\boldsymbol{\psi}) = \prod_{n=1}^{N} f(x_n; \boldsymbol{\psi}) = \prod_{n=1}^{N} \sum_{k=1}^{K} \lambda_k g(x_n; \boldsymbol{\theta}_k). \quad \text{where } \sum_{k=1}^{K} \lambda_k = 1. \tag{2}$$

One of the classic approaches for maximizing the likelihood function is to use an EM procedure in an iterative manner. Because the EM algorithm has been widely used in the field of computer vision and machine learning [50], we simply quote some previous result in this work.

Let $\boldsymbol{\theta}_k^{(m)}$ be the estimated parameter of the kth subcomponent in the mth iteration and w_{nk} be the soft assignment of pixel intensity x_n of each mixture subcomponent $g(x; \boldsymbol{\theta}_k)$ over the training intensity values. Then, the likelihood function can be maximized by iterating the following two steps.

1) **E-Step:** Compute the soft assignment w_{nk} of pixel x_n to each subcomponent $g(x; \boldsymbol{\theta}_k)$

$$w_{nk} = \frac{\lambda_k g(x_n; \theta_k)}{\sum_{j=1}^{K} \lambda_j g\left(x_n; \theta_j\right)} \tag{3}$$

for $k = 1, \ldots, K$ and $n = 1, \ldots, N$ using the current estimates of the parameters.

2) **M-Step:** Update parameters of the ΓMM using the posteriors computed in the E-Step. Especially, parameters $\boldsymbol{\psi}$ can be reestimated as

$$\hat{\lambda}_k = \frac{1}{N} \sum_{n=1}^{N} w_{nk}; \tag{4a}$$

$$\hat{\beta}_k = \frac{1}{\alpha_k} \left(\sum_{n=1}^{N} w_{nk} x_n / \sum_{n=1}^{N} w_{nk} \right); \tag{4b}$$

$$\begin{aligned} \nu(\hat{\alpha}_k) &= \log(\hat{\alpha}_k) - \Upsilon(\hat{\alpha}_k) \\ &= \log\left(\frac{\sum_{n=1}^{N} w_{nk} x_n}{\sum_{n=1}^{N} w_{nk}} \right) - \frac{\sum_{n=1}^{N} w_{nk} \log x_n}{\sum_{n=1}^{N} w_{nk}} \end{aligned} \tag{4c}$$

where $\Upsilon(x)$ is the *Digamma* function defined as $\Upsilon(x) = \Gamma'(x)/\Gamma(x)$.

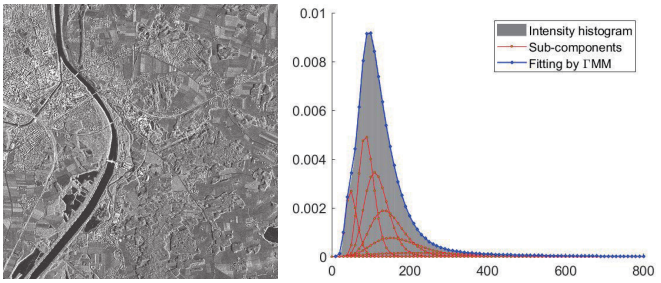
These two steps are repeated until the expected complete data log-likelihood ceases to improve or until a maximum number of iterations are reached.

As shown in 4, the update of weights is straightforward and is just the average value of the soft assignments computed in the previous step for each subcomponent. For shape parameter α_k, the update expression has no closed solution. However, it can be obtained by numerical methods since the function has the property

$$\nu(\hat{\alpha}_k) \approx \begin{cases} \dfrac{1}{2\hat{\alpha}_k} + \dfrac{1}{12\hat{\alpha}_k^2} + \cdots, & \text{as } \hat{\alpha}_k \to \infty \\ \dfrac{1}{2\hat{\alpha}_k} + \text{const}, & \text{as } \hat{\alpha}_k \to 0. \end{cases}$$

A simple gradient scheme should be sufficient to obtain the final solution. In addition, with the estimated shape parameter, the scale value β_k can also be updated directly according to 4.

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(a) (b)

Fig. 1. Histogram of SAR image fit by ΓMM with 20 subcomponents. (a) High-resolution SAR images acquired by TerraSAR-X satellite over Rosenheim. (b) Histograms and fitting by a ΓMM with 20-subcomponents. (a) SAR image. (b) Intensity histogram.

As an example, one high-resolution SAR image is selected to illustrate the effectiveness of the ΓMM for statistical modeling. Fig. 1(a) shows a high-resolution SAR image acquired by the TerraSAR-X satellite over Rosenheim, Bavaria, Germany. By fitting the intensity via a ΓMM with 20 subcomponents, the histogram and probability distributions are shown in Fig. 1(b). It can be observed that ΓMM can effectively characterize the high-peak and heavy-tailed properties of high-resolution SAR images. In Section III-A, the Gamma mixture model is utilized to capture the statistical characteristics for high-resolution SAR image feature extraction.

B. Fisher Vectors

In statistical classification, the Fisher kernel is widely used to measure the similarity of two objects on the basis of sets of measurements for each object and a statistical model. It combines the advantages of generative statistical models and those of discriminative methods, in which the generative models can process data of variable length, while the discriminative methods have flexible criteria and yield better results [41]–[43]. After fitting the probability density function of intensity values of high-resolution SAR images via a Gamma mixture model, the score function in statistics can be given by the gradient of the log-likelihood of the pixel intensities over the following model [41]:

$$G_{\boldsymbol{\psi}}^{x} = \nabla_{\boldsymbol{\psi}} \log f(x; \boldsymbol{\psi}). \tag{5}$$

This describes the contribution of individual parameters to the generative process of the SAR image. In other words, the score function measures how the parameter $\boldsymbol{\psi}$ is modified to better fit the statistical distribution of SAR intensities.

Then, the similarity between two pixels x_i and x_j can be defined using a Fisher kernel by combining the strengths of generative and discriminative models of these pixels. This is defined as follows:

$$s(x_i, x_j) = G_{\boldsymbol{\psi}}^{x_i \mathrm{T}} F_{\boldsymbol{\psi}}^{-1} G_{\boldsymbol{\psi}}^{x_j} \tag{6}$$

with a positive semidefinite Fisher information matrix $F_{\boldsymbol{\psi}}^{-1} \in \mathbb{R}^{M \times M}$, where M is the total number of parameters in ΓMM.

Using the Cholesky decomposition on the Fisher information matrix, i.e., $F_{\boldsymbol{\psi}}^{-1} = L_{\boldsymbol{\psi}}^{\mathrm{T}} L_{\boldsymbol{\psi}}$, the similarity $s(x_i, x_j)$ in 6 can be rewritten as

$$s(x_i, x_j) = G_{\boldsymbol{\psi}}^{x_i \mathrm{T}} L_{\boldsymbol{\psi}}^{\mathrm{T}} L_{\boldsymbol{\psi}} G_{\boldsymbol{\psi}}^{x_j} = \mathcal{G}_{\boldsymbol{\psi}}^{x_i \mathrm{T}} \mathcal{G}_{\boldsymbol{\psi}}^{x_j} \tag{7}$$

where

$$\mathcal{G}_{\boldsymbol{\psi}}^{x} = L_{\boldsymbol{\psi}} G_{\boldsymbol{\psi}}^{x} = L_{\boldsymbol{\psi}} \nabla_{\boldsymbol{\psi}} \log f(x; \boldsymbol{\psi}). \tag{8}$$

The normalized gradient term in 8 is called the FV for each pixel. It can be understood as an embedding of a single SAR pixel in a higher dimensional space that contains a large amount of generative information.

Equipped with ΓMM, a reparameterization of weight λ_k is constructed to avoid enforcing explicit constraints $\sum_{k=1}^{K} \lambda_k = 1$. This is defined as

$$w_k = \frac{\exp(\lambda_k)}{\sum_{j=1}^{K} \exp(\lambda_k)}. \tag{9}$$

The gradients of pixel x with respect to model parameter $\boldsymbol{\psi}$ of ΓMM can be calculated as

$$\nabla_{\boldsymbol{\psi}} \log f(p, \boldsymbol{\psi}) = \frac{\partial \log f(x; \boldsymbol{\psi})}{\partial \boldsymbol{\psi}} = \frac{1}{f(x; \boldsymbol{\psi})} \cdot \frac{\partial f(x; \boldsymbol{\psi})}{\partial \boldsymbol{\psi}}. \tag{10}$$

Especially, for each parameter λ_k, α_k, and β_k, the derivatives can be generated by

$$\frac{\partial f(x; \boldsymbol{\psi})}{\partial \lambda_k} = g(x; \boldsymbol{\theta}_k) \cdot \frac{\partial w_k}{\partial \lambda_k} = g(x; \boldsymbol{\theta}_k) \cdot w_k \cdot (1 - w_k) \tag{11a}$$

$$\frac{\partial f(x; \boldsymbol{\psi})}{\partial \alpha_k} = w_k \cdot \frac{\partial g(x; \boldsymbol{\theta}_k)}{\partial \alpha_k} = w_k \cdot g(x; \boldsymbol{\theta}_k) \cdot \left(\ln\left(\frac{x}{\beta_k}\right) + \frac{\Upsilon(\alpha_k)}{\Gamma(\alpha_k)} \right) \tag{11b}$$

$$\frac{\partial f(x; \boldsymbol{\psi})}{\partial \beta_k} = w_k \cdot \frac{\partial g(x; \boldsymbol{\theta}_k)}{\partial \beta_k} = w_k \cdot g(x; \boldsymbol{\theta}_k) \cdot \frac{x - \alpha_k \cdot \beta_k}{\beta_k^2} \tag{11c}$$

where $\Upsilon(\alpha_k)$ represents the Digmma function with respect to α_k, i.e., $\Upsilon(\alpha_k) = \Gamma'(\alpha_k)/\Gamma(\alpha_k)$.

Let

$$\gamma_k(x) = \frac{1}{f(x; \boldsymbol{\psi})} \cdot w_k \cdot g(x; \boldsymbol{\theta}_k) = \frac{w_k \cdot g(x; \boldsymbol{\theta}_k)}{\sum_{j=1}^{K} w_j g(x; \boldsymbol{\theta}_j)}$$

and the gradients of these parameters can be written as

$$\nabla_{\lambda_k} \log f(x, \boldsymbol{\psi}) = \gamma_k(x)(1 - w_k) \tag{12a}$$

$$\nabla_{\alpha_k} \log f(x, \boldsymbol{\psi}) = \gamma_k(x) \left(\ln\left(\frac{x}{\beta_k}\right) + \frac{\Upsilon(\alpha_k)}{\Gamma(\alpha_k)} \right) \tag{12b}$$

$$\nabla_{\beta_k} \log f(x, \boldsymbol{\psi}) = \gamma_k(x) \left(\frac{x - \alpha_k \cdot \beta_k}{\beta_k^2} \right). \tag{12c}$$

In addition, a diagonal Fisher information matrix $F_{\boldsymbol{\psi}}^{-1}$ is used to evaluate the projection matrix $L_{\boldsymbol{\psi}}$ under the assumption that the distribution of $\gamma_k(x)$ is sharply peaked on a single value k for any pixels. Therefore, the FV of each pixel x with respect to parameter $\boldsymbol{\psi}$ can be generated by a coordinatewise

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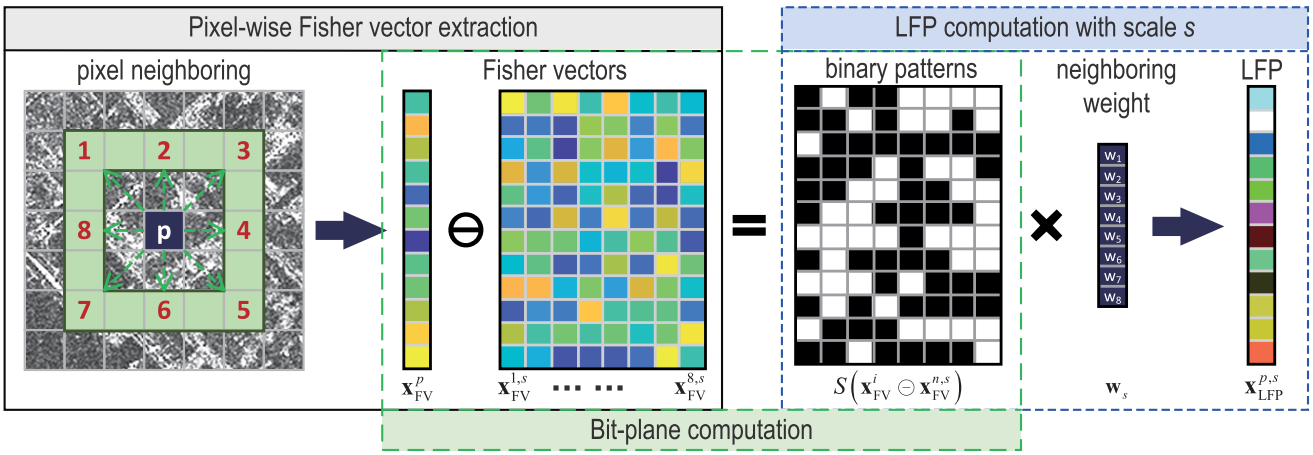


Fig. 2. Procedure for LFP computation.

normalization of gradient vectors in 12, which are listed as follows:

$$\mathcal{G}_{\psi}^{x} = \frac{1}{\sqrt{w_k}} \sum_{n=1}^{N} \nabla_{\psi} \log f(x_n; \psi). \tag{13}$$

As $\mathcal{G}_{\psi}^{x}$ is a pixelwise representation, its subcomponents, i.e., $\mathcal{G}_{\lambda_k}^{x}$, $\mathcal{G}_{\alpha_k}^{x}$, and $\mathcal{G}_{\beta_k}^{x}$, are scalars for $k = 1, \ldots, K$.

Therefore, the final FV for pixel x is constructed by concatenating all these gradients together and results in a feature vector

$$\mathbf{x}_{\mathrm{FV}} = \left\{\mathcal{G}_{\lambda_k}^{x}, \mathcal{G}_{\alpha_k}^{x}, \mathcal{G}_{\beta_k}^{x}\right\}_{k=1}^{K} \tag{14}$$

of $3 \times K$ elements in size. The approximation process of the Fisher information matrix can be found in [41].

C. Multiscale Local Fisher Pattern

To improve the representation capability of the pixelwise FVs, some spatial and geometrical constraints can be introduced to reduce the impact of speckles in the characterization of different land covers [51], [52]. Inspired by the work of LBP that is widely used to measure the local contrast in image texture analysis within a local neighborhood region, an MLFP is proposed to exploit rich local information between neighboring pixels.

Let $\mathbf{x}_{\mathrm{FV}}^{i}$ be the FV generated at the center pixel and $\mathbf{x}_{\mathrm{FV}}^{n,s}$ be FVs of its neighboring pixels within the neighborhood $Ngb(i,s)$ with scale s. Then, the local Fisher pattern (LFP) of the considered pixel i with specified scale value s can be defined as

$$\mathbf{x}_{\mathrm{LFP}}^{i,s} = \sum_{n \in Ngb(i,s)} \mathbf{w}_s S\left(\mathbf{x}_{\mathrm{FV}}^{i} \ominus \mathbf{x}_{\mathrm{FV}}^{n,s}\right) \text{ s.t. } S(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases} \tag{15}$$

where $\ominus$ represents the elementwise subtraction operation between vectors. The binary function $S(\cdot)$ constructs a bit plane by comparing FVs between the center pixel i and their neighbors $\mathbf{x}_{\mathrm{FV}}^{n,s}$. Meanwhile, $\mathbf{w}_s$ represents the vector for weighting each neighboring pixel in the generation of LFP value $\mathbf{x}_{\mathrm{LFP}}^{i,s}$.

As shown in Fig. 2, there are three major steps for LFP computation: 1) pixelwise FV extraction; 2) bit plane computation; and 3) LFP generation. Especially, in the first part, the FVs are extracted in a pixelwise manner for the whole SAR image. Then, a bit plane is generated for each pixel when comparing the considered pixel and its neighbors within scale s. Finally, a weight operator is applied to the generated bit plane to produce the LFP vector by considering the impact of its neighboring pixels.

With the learned LFP in 10, the MLFP vector can be generated by concatenating LFPs $\mathbf{x}_{\mathrm{LFP}}^{i,s}$ with S different scales, $scl = (s_1, s_2, \ldots, s_S)$, which can be described as

$$\mathbf{x}_{\mathrm{MLFP}}^{i} = \left(\mathbf{x}_{\mathrm{LFP}}^{i,s_1}, \ldots, \mathbf{x}_{\mathrm{LFP}}^{i,s_S}\right) \tag{16}$$

where $\mathbf{x}_{\mathrm{LFP}}^{i,s_i}$ denotes the LFP vector computed for pixel x^i with scale value s_i. Then, each pixel can be characterized by a feature vector concatenating its intensity value x^i, the FV $\mathbf{x}_{\mathrm{FV}}^{i}$, and MLFPs $\mathbf{x}_{\mathrm{MLFP}}^{i}$ together, i.e., $\mathbf{x}^i = \left(x^i, \mathbf{x}_{\mathrm{FV}}^{i}, \mathbf{x}_{\mathrm{MLFP}}^{i}\right)$. The pseudocode for the pixelwise feature vector utilized in this work is shown in Algorithm 1.

Algorithm 1 MLFP Vector Computation

Input: Training SAR intensity values set, number of mixture components K and scale values $s_1, s_2, \ldots, s_S$.
Output: Feature vector $\mathbf{x}^i$ for each pixel.
1: Fitting a K-components Gamma mixture model over the training intensities set via an EM iteration procedure in Eq. 3 and Eq. 4;
2: Compute gradients of each pixel with respect to model parameters in Eq. 12, and concatenate them to formulate Fisher vector $\mathbf{x}_{\mathrm{FV}}$;
3: **for** scale s in $\{s_1, s_2, \ldots, s_S\}$ **do**
4: Compute local Fisher pattern vector $\mathbf{x}_{\mathrm{LFP}}^{i,s}$ in Eq. 15;
5: **end for**
6: Concatenate the local Fisher pattern vectors with different scales together as declared in Eq. 16;
7: **return** Pixel-wise feature vector: $\left(x^i, \mathbf{x}_{\mathrm{FV}}^{i}, \mathbf{x}_{\mathrm{MLFP}}^{i}\right)$.

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The major computational burden of Algorithm 1 lies in the fitting procedure of ΓMM, which is heavily related to the number of subcomponents and the number of intensity values for model training. This may be terminated with a specified number of iterations or a certain tolerance for probability approximation. However, with the learned ΓMM, the FVs and MLFPs can be calculated mathematically in 14 and 15. This is a simple and different way to characterize different pixels.

D. Discussions of Proposed Approach

In this section, we discuss the proposed MLFP. First, the statistical properties of high-resolution SAR image intensities can be effectively characterized by ΓMM, which is sufficient to model the probability density functions of different land covers. After the introduction of the geometrical information between neighboring pixels within a certain scale, the proposed MLFP can effective exploit the rich generative and discriminative information being contained in the FVs. Meanwhile, the multiscale spatial structure information is also exploited by MLFP for each pixel to characterize different local structures of high-resolution SAR textures, which is critical for distinguishing different land covers. Then, compared with the classic mid-level and deep features, the MLFP is a shallow representation of high-resolution SAR images. Therefore, less runtime is required for feature extraction in the characterization of different land covers without any extra model training. However, this can also be utilized as the basic feature to learn a large amount of abstract and higher level features within some classic frameworks after the introduction of certain prior knowledge, such as bag-of-words, sparse representation, and stacked autoencoder (SAE). Finally, as one of the pixelwise representations, some preprocessing or postprocessing tricks should be adopted to improve the computational efficiency in the use of the proposed MLFP to label different land covers, such as superpixels, patchwise processing, and orphan pixel processing [53], [54]. In this work, we evaluate the performance of the proposed MLFP on two experiments: high-resolution SAR image land-cover classification and SAR scene category. For land-cover classification, superpixels are employed as a preprocessing technique for reducing the computational cost.

III. Experimental Results and Discussion

To evaluate the performance of the proposed approach, three experiments are conducted in this section. First, a land-cover classification problem is considered on one selected high-resolution SAR image. Then, we consider a high-resolution SAR scene category problem in the second part of this experiment. Finally, some discussions are proposed in the last part of this section.

In this experiment, several feature descriptors that are widely utilized in the field of SAR image processing are adopted for performance comparison. Especially, they are listed as follows.

1) *Patch Vector:* In this experiment, the neighboring patch vector of each pixel is utilized to exploit its spatial neighborhood characteristics as the benchmark for performance comparison.
2) *Textures [55]:* In addition, some textures that are generated from GLCM are also employed to characterize different land covers. Especially, a neighboring patch of 11×11 pixels in size is selected in the first stage for the considered pixel. Then, the GLCM is generated with degrees $\{0, 90, 180, 270\}$ by requantizing pixel intensity values into 1024 levels. The final textures that are used to characterize different land covers in this experiment are contrast, correlation, energy, and homogeneity computed from each GLCM, which results in a feature vector with 16 elements.
3) *Gabor Filters [15]:* In addition to a patch vector and textures, the mean and standard deviations of the magnitude of the Gabor filter response with scales {3, 5, 7, 9, 11, 13, 15, 17} and orientations {0, 30, 60, 90, 120, 150} are employed to characterize each pixel in this experiment. This results in a feature vector with 96 elements, in which the former 48 elements are means and the latter 48 are standard deviations.
4) *Bag-of-Words (BoW) [56]:* To generate the bag-of-words representation of SAR images, each pixel is characterized by its neighborhood patch vector with size $[11 \times 11]$ in the first stage. Then, a codebook with 128 elements is generated from all patch vectors via a K-means clustering procedure. After the number of all codewords that appear is counted, the SAR image can be characterized by a feature vector $\mathbf{f} = (f_1, f_2, \ldots, f_{128})$ with 128 elements, in which f_d denotes the normalized frequency of the dth-codeword.
5) *SAE [57]:* In this experiment, an SAE with three hidden layers is constructed for each neighborhood patch with size $[11 \times 11]$ of the considered pixel. Especially, the network structure and parameter settings of SAE used in this experiment are shown in Fig. 3. This results in a feature vector for each pixel with 100 elements.

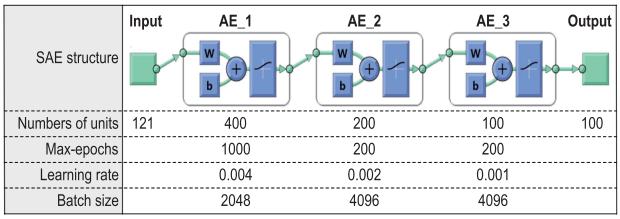


Fig. 3. Network structure and parameter settings of SAE in this experiment.

Before feeding to a classifier, all features are normalized to have zero-mean and unit variance.

In addition, four metrics, i.e., accuracy, precision, recall, and f1-score, which are widely used in the field of machine learning, are adopted to evaluate the performance of each feature vector. These are listed in Table I. Especially, accuracy is utilized to measure the overall discriminant capability of different features. A higher value indicates that the feature has excellent discrimination capability when characterizing different land covers. Precision is the proportion of

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TABLE I
METRICS FOR CLASSIFICATION PERFORMANCE EVALUATION

metrics	accuracy	precision	recall	f1-score
explain	the percentage of corrected classification samples for each category	the fraction of correct classified samples among all labeled samples	the fraction of the total amount of true instances that were actually labeled	an overall measure of the classifier's accuracy by combing precision and recall
computation	$accuracy = \frac{TP+TN}{TP+TN+FP+FN}$	$precision = \frac{TP}{TP+FP}$	$recall = \frac{TP}{TP+FN}$	$f1 = \frac{2 \times precision \times recall}{precision + recall}$

* To evaluate each metric, TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. Details of these values can be found in [59].

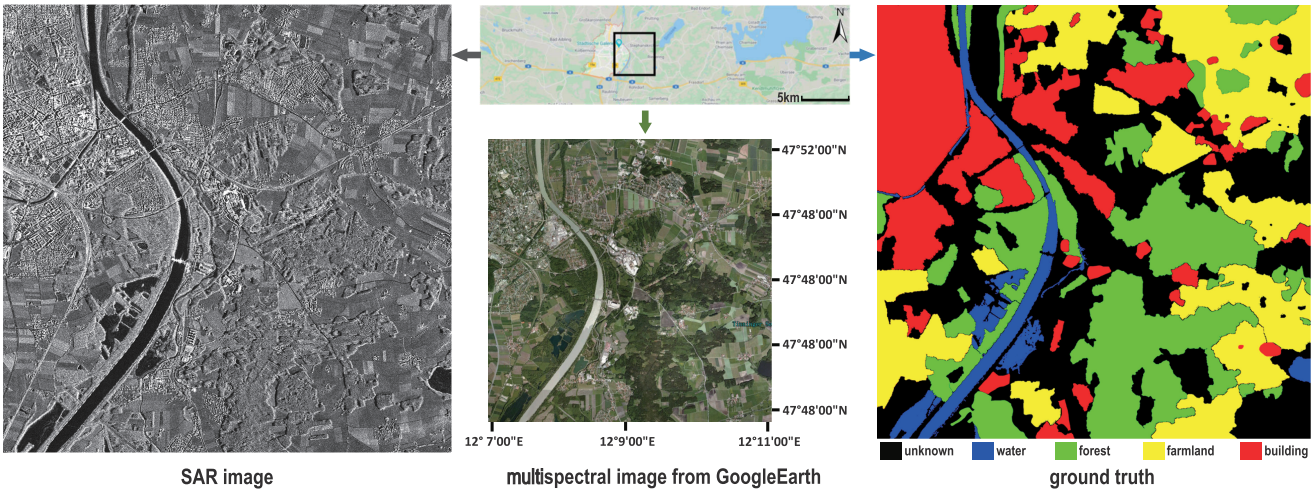


Fig. 4. (Left) Selected SAR image fragment. (Middle) Referenced multispectral image from the geospatial location. (Right) Ground truth of different land covers of selected SAR image.

correctly labeled samples among all positive predictions, while recall measures the probability that a classifier assigns a sample to its expected category. In land cover classification task, the recall value reflects the capability of the proposed approach to identify positive samples, while the precision value reports the distinguishing capability of negative samples. After combing precision and recall values together, a higher f1-score implies that the proposed approach has much robust discriminant capability while distinguishing different land covers. As these metrics are used for two-category classification problems, they are evaluated in a one-versus-the-rest manner for each land cover when the number of categories that should be distinguished is larger than two in the following experiments. In addition to these metrics, the overall accuracy (*o.a.*) and Cohen's kappa coefficient (κ) [58] are computed to evaluate the overall performance of different approaches. In these experiments, a radial bias function (RBF) kernel support vector machine (SVM)[1] is trained to evaluate the discriminant capability of these features [60]. All experiments are conducted on a single PC with Intel core I7-8700K CPU and 32-GB memory and implemented over MATLAB 2019a. Among these features, the network of SAEs is trained on an Nvidia GTX 1080 GPU, while the others are carried out in a single-CPU environment.

A. High-Resolution SAR Image Land-Cover Classification

1) Data Sets and Experimental Settings: In this experiment, a high-resolution SAR image that was acquired over Rosenheim (see Fig. 4) by the TerraSAR-X satellite with an horizontal transmit and horizontal receive (HH)-polarization spotlight mode is employed to evaluate the discriminant capability of the proposed approach. After two-look preprocessing, a fragment shown in the left part of Fig. 4 with 3600 × 3600 pixels is selected from the original SAR image for the land-cover classification problem. Four land covers can be funded in the considered SAR image according to the multispectral image from GoogleEarth (as shown in the middle part of Fig. 4): water, forest, farmland, and buildings. According to the multispectral image, the ground truth of the selected SAR image is labeled manually and shown in the right part of Fig. 4.

To effectively label different land covers, superpixels are segmented and utilized on the selected SAR image in the first place. Especially, the simple linear iterative clustering (SLIC)[2] superpixel segmentation approach is utilized in this experiment due to its simplicity and capability to maintain

[1]Code available from: http://www.csie.ntu.edu.tw/∼cjlin/libsvm/. In all experiments, the Gamma value in RBF kernel function is setting as 0.07 for model training without shrinking heuristics.

[2]Code available from: http://www.epfl.ch/labs/ivrl/research/slic-superpixels

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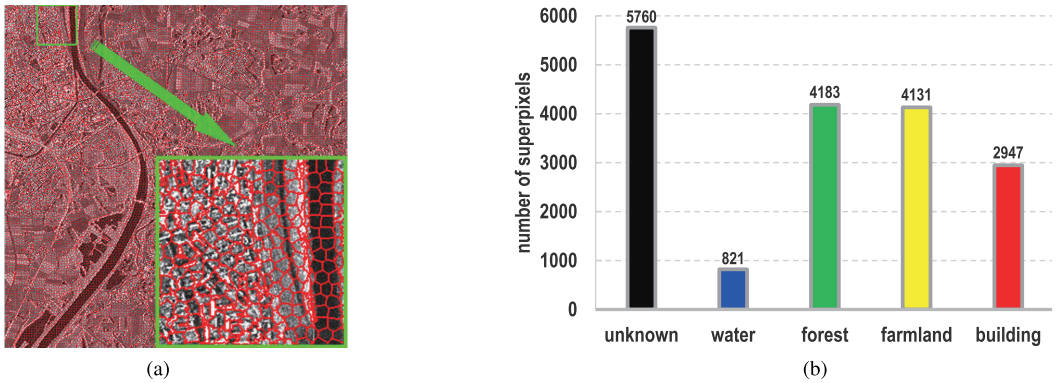


Fig. 5. Superpixel segmentation and number of labeled superpixels for classification. (a) Superpixels generated by SLIC. (b) Plots of the number of labeled superpixels from different land covers. (a) SLIC superpixels. (b) Number of labeled superpixels.

the boundaries [61]. For a high-resolution SAR image, the distance between pixel i and j is defined as [62]

$$\text{dist}(i,j) = \sqrt{\lambda\left(\frac{|x_i+x_j|}{\sqrt{x_i \cdot x_j}}\right)+(1-\lambda)\left((a_i-a_j)^2+(b_i-b_j)^2\right)}$$

where (a_i, b_i) and (a_j, b_j) are the spatial coordinates of pixels i and j, respectively. x_i and x_j represent the corresponding intensity values of each pixel. λ is the regularization parameter that takes a balance between the spatial location and intensity values in the comparison of these two pixels. Given this distance, the generated superpixels are shown in Fig. 5(a).

With the labeled land covers of the selected SAR image, more than 60% of the pixels were labeled previously in the referenced ground truth image. Consequently, 821, 4183, 4131, and 2947 superpixels for water, forest, farmland, and built-up areas can be drawn and used for land-cover classification in this experiment. In addition to the labeled superpixels, there are 5760 superpixels without any label information. These may be radar shadows and the other undefined land covers. The numbers of superpixels from different land covers in this experiment are plotted in Fig. 5(b).

2) Analysis of Results: In this experiment, 50% of the labeled superpixels are randomly selected for classifier training, while the remaining 50% of the superpixels are utilized for testing. After this experiment was conducted 20 times independently, one classification result of the selected SAR image is selected randomly from different feature vectors and plotted in Fig. 6, respectively.

It can be observed that the classification results for the patch vector [see Fig. 6(a)] and textures [see Fig. 6(b)] appear to have more clutter than those obtained from the other features. For the patch vector, a large amount of misclassification occurred between superpixels from building and farmland areas. Numerous superpixels from these two land covers are mislabeled as a forest because it is difficult to capture enough discriminant information by only employing the intensities from the neighborhood. Because the textures used in this experiment are generated from the GLCM of a local SAR image patch, this method cannot effectively capture rich and accurate statistical information of different land covers. Therefore, many superpixels of the building area are mislabeled as forest. Furthermore, superpixels from water areas in the selected SAR image fragment are not marked clearly because the number of training and testing samples is very small compared with that of the others. For features generated from the Gabor filters [see Fig. 6(c)], this method can effectively distinguish superpixels from farmland and built-up areas. This may be caused by the fact that the local structures of these two land covers appear more regularly and are largely man-made targets. As there is not significant local structure information, almost all superpixels from water areas are mislabeled as farmland for the Gabor filters. Significant improvement can be observed from BoW [see Fig. 6(d)], SAE [see Fig. 6(e)], and the proposed MLFP [see Fig. 6(f)]. Because BoW and SAE can effectively capture mid-level, high-level, and abstract representations, many misclassifications of superpixels occur for these two features. However, numerous superpixels from the water area are mislabeled for SAE, and the number of superpixels is much lower than those of the other three land covers. It is difficult to train an effective network with fewer samples to obtain effective features. The proposed MLFP outperforms the other five feature vectors because it can simultaneously capture statistical information and structural information of different land covers, specifically for superpixels from water, building, and forest areas.

Apart from the visualized classification results of different feature vectors, the average overall accuracy (*o.a.*) and Cohen's kappa coefficient (κ) are recorded and listed in Table II. It can be observed that the proposed MLFP obtains the best values of *o.a.* and κ. In addition, the average times of the major steps of these approaches are recorded and listed in Table II, which are feature extraction, SVM model training, and SVM prediction. Compared with SVM model training and prediction, a large amount of runtime is needed for feature extraction except for the patch vector. Among all these features, the runtime for patch vector extraction is minimal because it just partitions the image into several neighbor patches without any further processing. It can be observed that a large amount of runtime is needed for texture extraction because a very large number of GLCMs should be generated

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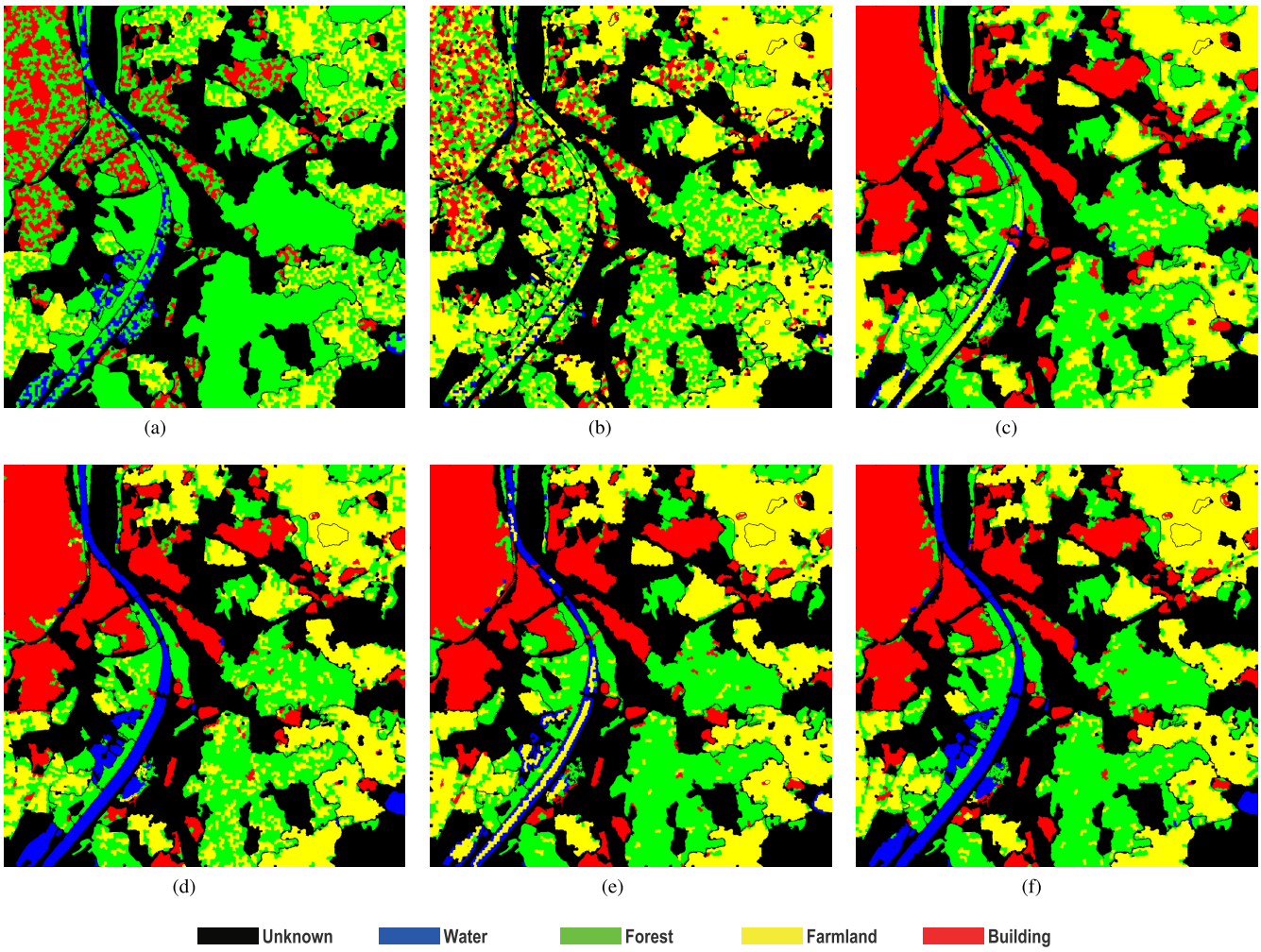


Fig. 6. Classification results for selected SAR image fragment for (a) patch vector, (b) textures, (c) Gabor filters, (d) BoW, (e) SAE, and (f) MLFP.

TABLE II
AVERAGE OVERALL ACCURACY (*o.a.*), COHEN'S KAPPA COEFFICIENT (κ), AND RUNTIME OF DIFFERENT FEATURES

	feature vectors	patch-vector	textures	Gabor filters	BoW	SAE*	MLFP
	o.a. (100%)	65.55 ± 0.46	57.76 ± 0.45	78.21 ± 0.15	82.20 ± 0.08	83.22 ± 0.21	**88.33** ± 0.06
	κ	0.5182 ± 0.0072	0.3336 ± 0.0050	0.7049 ± 0.0025	0.7825 ± 0.0019	0.8057 ± 0.0030	**0.8746** ± 0.0013
Runtime(s)$^+$	***ft. ext.***	21.88 ± 0.01	7.71E4 ± 8.32E2	43.83 ± 3.21	1891.95 ± 35.23	943.66 ± 19.80	373.70 ± 15.83
	SVM tr.	68.94 ± 12.01	10.84 ± 0.25	6.71 ± 0.16	6.45 ± 0.08	12.03 ± 0.40	12.20 ± 0.18
	SVM pred.	25.01 ± 3.74	3.49 ± 0.04	3.90 ± 0.07	5.97 ± 0.13	6.59 ± 0.15	6.24 ± 0.51

* Runtime for SAE containing deep network training, which is conducted on an Nvidia GTX 1080 GPU for acceleration. + *ft. ext.* for feature extraction, *SVM tr.* for SVM model training, and *SVM pred.* for SVM prediction.

in a pixelwise manner. Especially, an SAR image with 3600 × 3600 pixels may generate more than 12M neighbor patches, which may consume the greatest amount of CPU computing resources for GLCM and texture generation. For BoW, a large amount of runtime is required for codeword generation via K-means clustering. Compared with that required for the other methods, the runtime required for MLFP feature extraction is moderate, which may be due to the statistical modeling of intensity values in the fitting of the Gamma mixture model.

To precisely quantify the classification performance, the confusion matrices (also known as error matrices) of different feature vectors are computed and shown in Fig. 7. Especially, each row of the confusion matrix represents the instances in a predicted category, while each column represents the instances in an actual class label of different land covers. Because it is easy to see if the classification system is confusing different land covers, the confusing matrix is a useful tool to evaluate the performance of different approaches. From the visualized result shown in Fig. 7, a similar conclusion can be obtained as the former part. Among all these approaches, the classification result of BoW [see Fig. 7(d)] and the proposed MLFP [see Fig. 7(f)] are acceptable, as fewer misclassifications occur. For superpixels from water areas,

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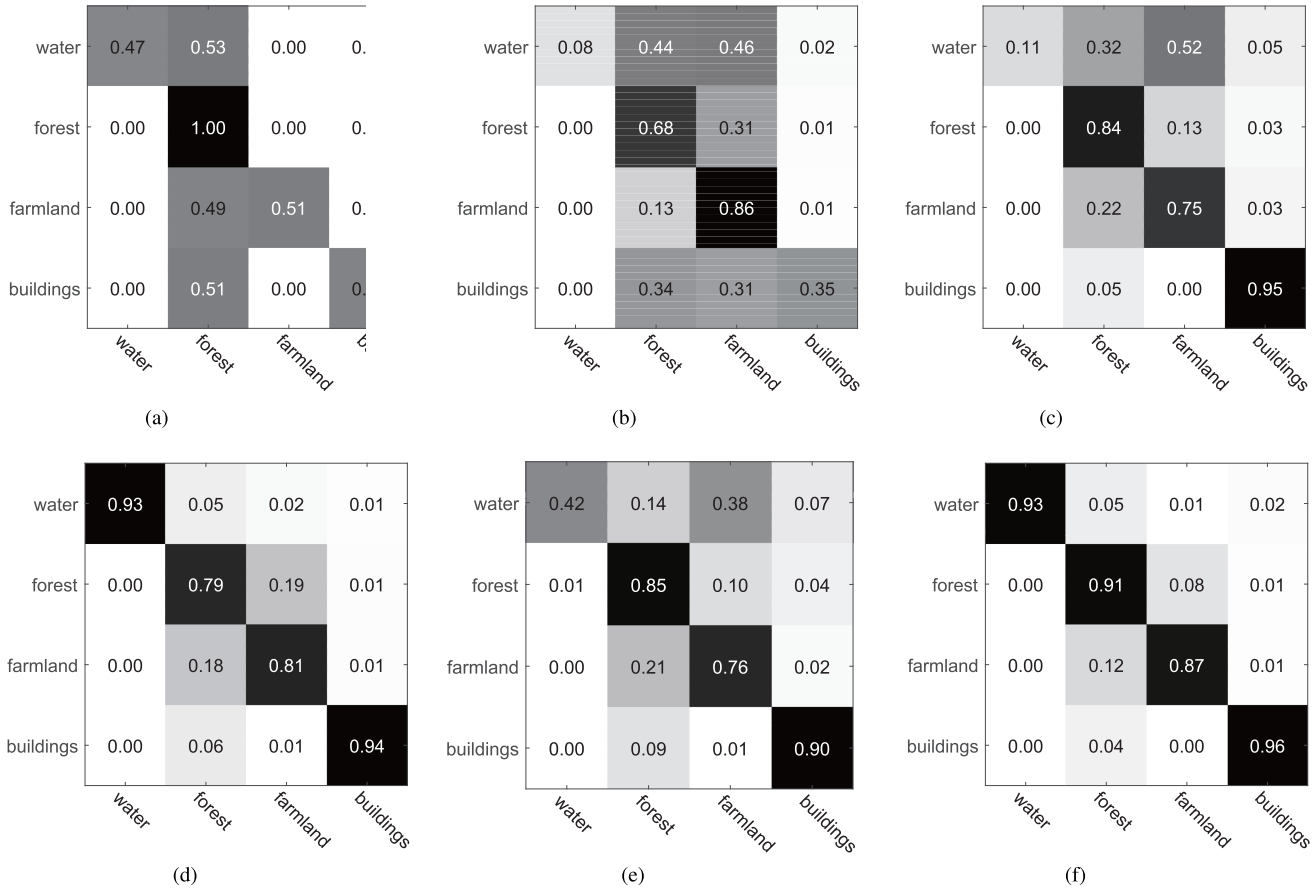


Fig. 7. Confusion matrix of different feature descriptors over selected SAR image fragments for (a) patch vector, (b) textures, (c) Gabor filters, (d) BoW, (e) SAE, and (f) MLFP.

misclassification is more likely to occur for patch vectors [see Fig. 7(a)], textures [see Fig. 7(b)], the Gabor filters [see Fig. 7(c)], and SAE [see Fig. 7(e)], in which they are mislabeled as forest or farmland. Compared with those in the proposed MLFP, many misclassifications occur between forests and farmland as a result of similar spatial structures between these two land covers.

Finally, four metrics, i.e., accuracy, precision, recall, and f1-score, are also computed for different land covers with different feature vectors. These are shown in Fig. 8. They are utilized widely to evaluate the advantages of a two-class classifier. In this experiment, these metrics are computed by casting a four-category classification problem as four two-category classification problems in a one-versus-the-rest manner for each different land cover. It can be observed that the accuracy [see Fig. 8(a)] and precision [see Fig. 8(b)] values of superpixels from water areas are high for all features but have an obvious difference over recall [see Fig. 8(c)] and the f1-score [see Fig. 8(d)]. This may be caused by the fact that, for superpixels in water areas, the classification problem is extremely imbalanced and almost reaches 1:14 in this experiment. Superpixels from forests, farmland, and built-up areas have higher values for these four metrics (higher than 80%).

B. Scene Classification

Apart from the high-resolution SAR image land-cover classification, distinguishing different scenes is another important application of SAR image understanding, especially with the increase in the ground resolution. In this experiment, we will evaluate the classification performance of the proposed approach. Especially, this evaluation includes two parts: scene data set construction and performance comparison.

1) Data Set and Parameter Settings: Similar to the work of [63], [64], the scene data set used in this experiment is constructed by applying a regular grid to several high-resolution SAR images that are acquired by the TerraSAR-X satellite in HH polarization with spotlight mode. These are imaged over the Rosenheim, Toronto, Java, Colorado, Beijing, and Hong Kong airports. Rich scenes can be found in these images, such as large-scale farmland and forest areas, water surfaces, and residential areas with different densities. Especially, as shown in Fig. 9(a), the scene data set is constructed by tiling the regular grids (with 200×00 pixels in size) on all images with 50% pixel overlap in both horizontal and vertical directions in this experiment. The final scene set consists of 10 000 patches with 25 different scene categories, each with 400 samples. Some samples of the constructed scene

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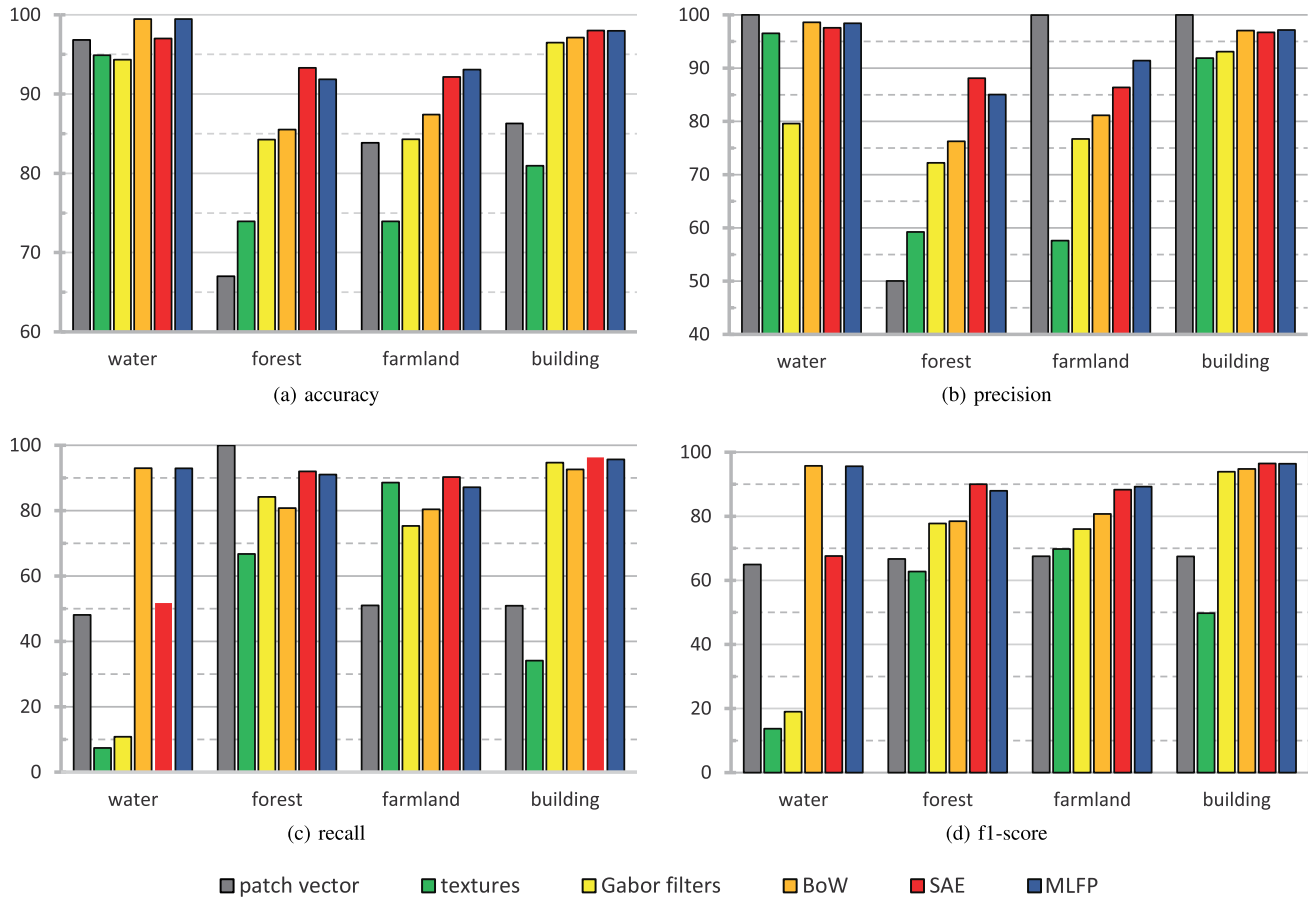


Fig. 8. Metrics of different feature vectors when distinguishing different land covers for (a) accuracy, (b) precision, (c) recall, and (d) f1-score.

database are shown in Fig. 9(b). It can be observed that some scene categories, such as C01, C02, C03, C05, and C11, contain large homogeneous regions for bare land, water, and large-scale farmland. On the other hand, scenes C08, C09, C10, C15, and C16 exhibit particular scatter effects due to different types of man-made objects. Scenes C11, C12, C13, C14, and C15 are samples of residential areas with different densities.

Like experiments in Section III-A, the patch vectors, textures, the Gabor filters, BoW, and SAE are utilized as comparison methods of the proposed MLFP with the same settings. Especially, an RBF-kernel-based SVM is used to classify different scene images with different feature vectors. In this experiment, 50% of the scene samples are used for SVM model training and the other 50% of the samples for performance testing in each category. After this experiment was conducted ten times independently, an analysis of the proposed approach was conducted and is discussed in the following subsection.

2) Analysis of Results: First, the average accuracy, precision, recall, and f1-score values are also computed and recorded after reformulating the proposed multiclass problem as several binary classification problems for each category in a one-versus-the-rest manner. These are shown in Fig. 10. For the accuracy values [see Fig. 10(a)], it can be observed that more than 17 categories obtain the highest values for the proposed MLFP. This indicates that MLFP can effectively label different scene images by exploiting both the statistical and spatial information contained in different scene samples. Meanwhile, for the precision [see Fig. 10(b)] and recall [see Fig. 10(c)] values, it can also be observed that the proposed MLFP obtains the highest values over the gross scene categories. By considering the precision and recall values at the same runtime, the f1-scores [see Fig. 10(d)] of the proposed MLFP reflect the advantages compared with the other feature vectors. The proposed MLFP has the highest f1-values over 20 categories. In addition, it can also be observed that textures have superior performance than in the former experiment.

Second, in order to precisely check the performance over different scenes, one confusion matrix of the proposed MLFP from this experiment is selected and shown in Fig. 11. It can be observed that there are 13 categories with a precision value higher than 0.95: C01, C02, C03, C05, C06, C07, C08, C09, C10, C14, C15, C19, and C21. This may be because the spatial structures in these scenes are simple and obvious to capture by the proposed MLFP. However, the classification performance of scenes C11, C18, and C25 is even poorer and lower than 0.75 in this experiment. It can be observed from the samples in Fig. 9(b) that the samples from scenes C11 and C25 have similar structures, and it is difficult to distinguish them. From the confusion matrix, it can be observed

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(a) (b)

Fig. 9. Procedure of SAR scene data set construction (a) for patch extraction and (b) displaying samples from each category. Especially, C01 for bare land; C02 for the runway; C03, C04, and C05 for water area; C06 for aquaculture; C07, C08, C09, and C10 for different man-made buildings; C11, C12, C13, C14, and C15 for residential areas of various densities; C16 for skyscrapers; C17 and C18 for rivers; C19 for freeways; C20 for roads in city area; C21 for large-scale farmland; C22 for the forest; C23 for a mixture of residential area and farmland; C24 for train tracks; and, finally, C25 for a mixture of tall vegetation and farmland. (a) Patch extraction. (b) Data set.

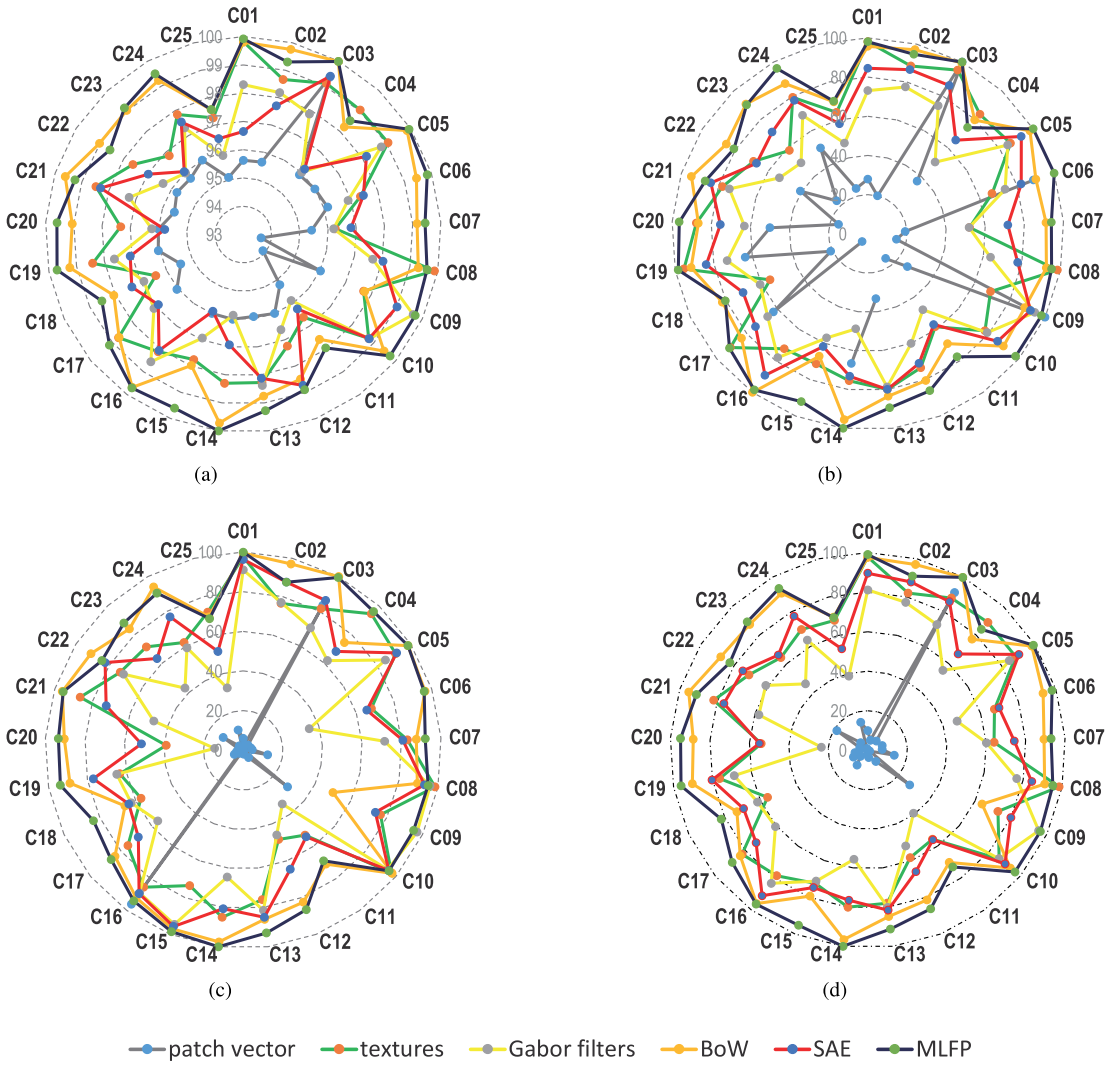


Fig. 10. Metrics of different feature vectors for SAR scene classification for (a) accuracy, (b) precision, (c) recall, and (d) f1-score.

that almost 20% of the samples from C11 are misclassified as C25. Meanwhile, 14% of the samples from scene C25 are misclassified as C11. This indicates that the samples from scenes C11 and C25 are the most difficult to distinguish from each other in this experiment. In addition, the samples in scene C18, i.e., rivers crossing artificial buildings, are likely mislabeled as C7 (man-made buildings), C12 (man-made buildings), and C24 (train tracks) because they largely have

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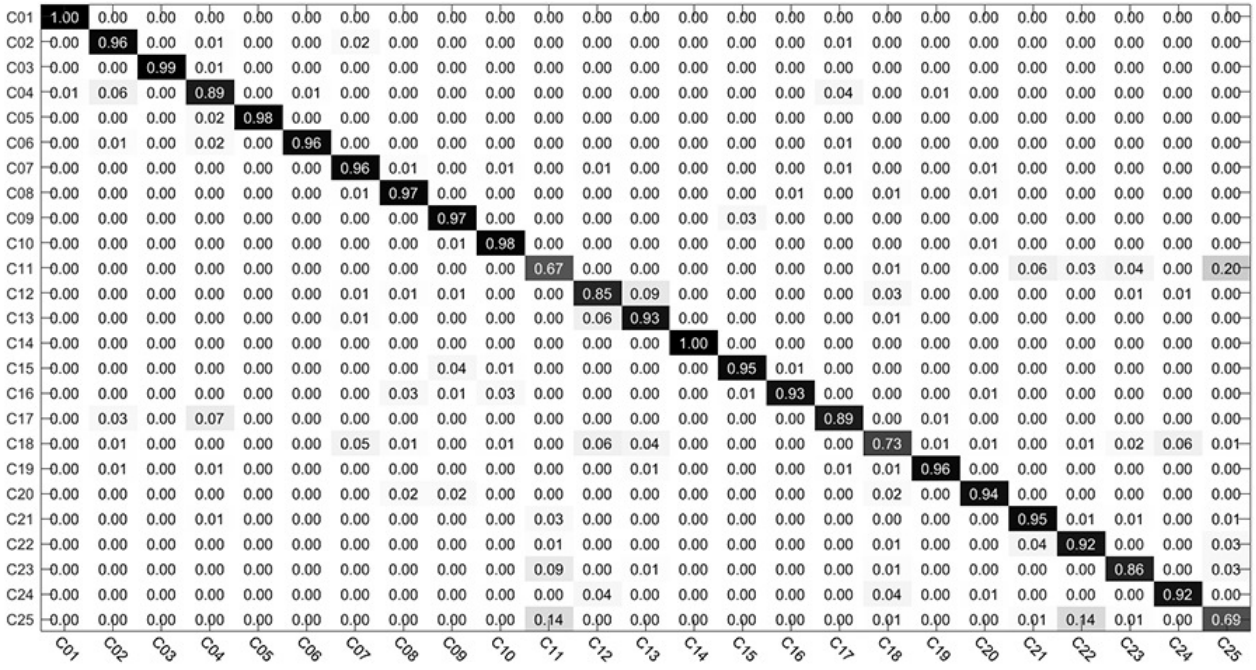


Fig. 11. Confusion matrix of proposed MLFP for SAR scene classification.

TABLE III

AVERAGE OVERALL ACCURACIES (*o.a.*), COHEN'S KAPPA COEFFICIENTS (κ), AND RUNTIMES OF DIFFERENT APPROACHES FOR HIGH-RESOLUTION SAR IMAGE SCENE CLASSIFICATION

Approaches	*o.a.* (%)	κ	*Runtime*(s)$^+$		
			ft. ext.	*SVM tr.*	*SVM pred.*
patch vector	9.44 ± 1.41	–	62.49 ± 4.90	35.31 ± 2.27	4.31 ± 0.30
textures	75.03 ± 0.49	0.7399 ± 0.0051	256.20 ± 16.93	3.65 ± 0.13	0.99 ± 0.01
Gabor filters	65.17 ± 0.40	0.6372 ± 0.0042	1706.34 ± 40.98	6.80 ± 0.09	1.81 ± 0.05
BoW	87.28 ± 0.22	0.8675 ± 0.0023	1134.41 ± 66.78	7.89 ± 0.37	4.84 ± 0.45
SAE*	85.36 ± 0.42	0.8231 ± 0.0037	1125.95 ± 89.35	12.05 ± 0.83	5.32 ± 0.34
MLFP	**91.10** ± 0.20	**0.9072** ± 0.0019	614.05 ± 58.97	11.77 ± 2.66	1.93 ± 0.13

* Runtime for SAE contains deep network training, which is conducted on an Nvidia GTX 1080 GPU for acceleration: + *ft. ext.* for feature extraction, *SVM tr.* for SVM model training and *SVM pred.* for SVM prediction.

similar structures. For these scenes, it is a challenging task for humans to visually interpret radar shadows, foreshortening, layover, and occlusion effects that widely appear in high-resolution SAR images.

Finally, the average overall accuracy (*o.a.*), Cohen's kappa coefficient (κ), and runtime are recorded and listed in Table III. It can be observed that the proposed MLFP has the best performance over the other approaches. Meanwhile, the times of the three major steps of scene classification are recorded. Compared with the Gabor filtering, SAE, and BoW, the proposed MLFP requires significantly less runtime for feature extraction. This may be because the computational cost for MLFP for each scene sample is much lower than those of the other approaches with an existing ΓMM. Nevertheless, the runtime of fitting an appropriate statistical model is not heavily dependent on the number of pixels from scene samples. As the textures are computed over the whole SAR scene sample image, this method needs much less runtime than the other experiments.

C. Discussion

In this section, three experiments are conducted to discuss the proposed approach: parameter analysis, sensitivity to different numbers of training samples, and sensitivity to classifier selection. As in Section III-A, a superpixelwise land-cover classification problem over a Rosenheim image (as shown in Fig. 3) is considered in this experiment to evaluate the performance with respect to different settings of the proposed approach.

1) Parameter Analysis: There are two important parameters of the proposed MLFP approach that should be determined: 1) the number of subcomponents in ΓMM and 2) the scale values for MLFP generation. Especially, we first check the performance of the ΓMM for high-resolution SAR image statistical distribution modeling. Then, the classification performance is checked with different settings of these two parameters.

First, we check the performance of the Gamma mixture model to approximate the statistical distribution of

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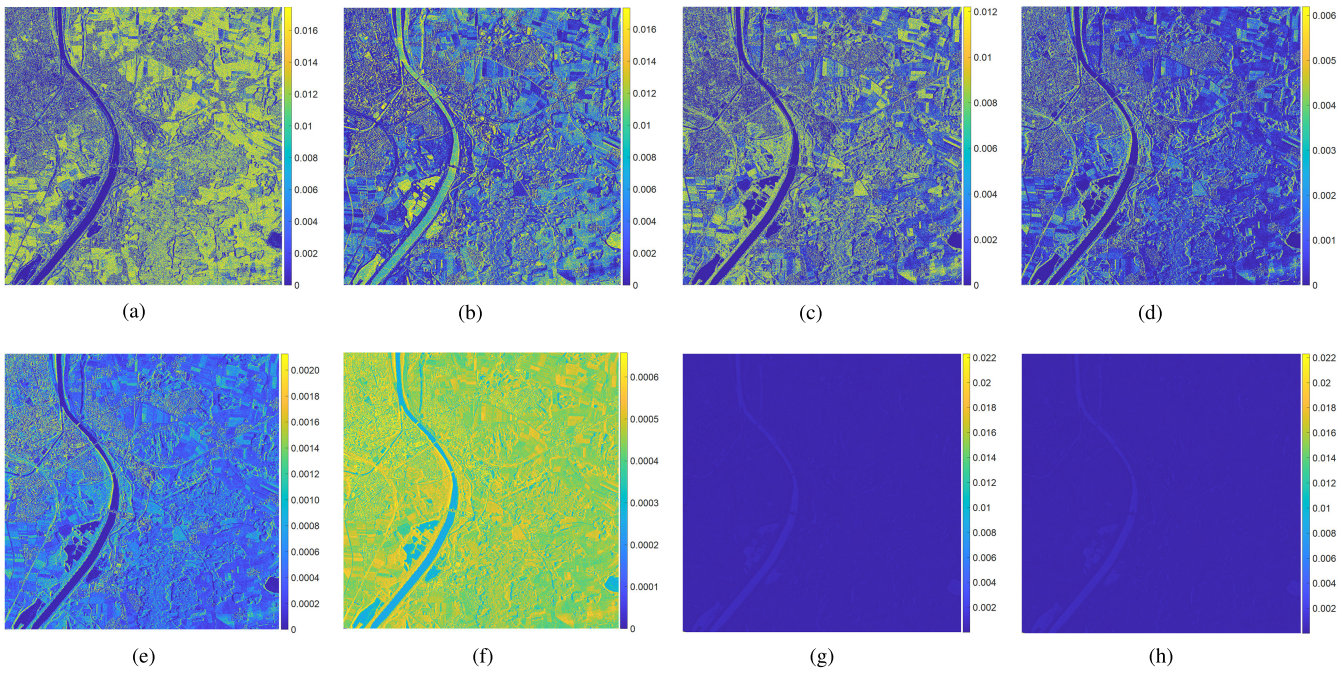


Fig. 12. Visualized probability response of the Rosenheim image with respect over different subcomponent of the ΓMM. (a) First subcomponent. (b) Second subcomponent. (c) Third subcomponent. (d) Fourth subcomponent. (e) Fifth subcomponent. (f) Sixth subcomponent. (g) Seventh subcomponent. (h) Eighth subcomponent.

high-resolution SAR image data. Especially, the number of mixture components of the Gamma mixture model is setting as 8. The visualized probability values of the Rosenheim image pixels with respect to each subcomponent of ΓMM are shown in Fig. 12, respectively. It can be found that the statistical distributions of different land covers could be effectively modeled by these subcomponents of ΓMM. As shown in Fig. 12(a), it can be observed that the farmland area has a relatively large probability response over the first subcomponent than the other land covers. In Fig. 12(b), the water area has a high probability value over the second subcomponent of ΓMM. Some forest areas can be clearly observed from Fig. 12(c), especially the ones along the edge of the river in the Rosenheim image. Meanwhile, the strong scatter response that came from building areas can also be detected in Fig. 12(d) and (e). However, a large amount of wrong responses was appeared between different land covers over these subcomponents, especially in building and forest areas. This is caused by the fact that ΓMM used in this article is a pixelwise statistical model without importing any spatial information implied by different land covers. In addition, it can be observed that the SAR image has minimal probability response values on the last two subcomponents [as shown in Fig. 12(g) and (h)], which is caused by the fact that these two subcomponents are generated to reduce the fitting error between ΓMM and the statistical distribution of real high-resolution SAR image.

Second, we check the classification performance with respect to different numbers of subcomponents in the Gamma mixture model. In this experiment, 50% of the superpixels from the Rosenheim image are selected randomly for SVM model training, while the remaining 50% are utilized for model testing. The numbers of subcomponent in ΓMM are setting as $K = \{2, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, \text{and } 60\}$, respectively. After the experiment is conducted ten times independently, the average overall accuracies (*o.a.*) and Cohen's kappa coefficients (κ) are listed in Table IV when setting the scales as $\text{scl} = [1, 3, 5]$. It can be observed that the highest *o.a.* is reached with 12 subcomponents of the ΓMM. Meanwhile, it can also be observed that, with a mixture number higher than 8, the average *o.a.* of the proposed approach is higher than 88%, and its performance does not change substantially with respect to an increasing number of mixture components.

In addition to the average *o.a.* and κ, the average times for the Gamma mixture model training, FV computation, MLFP generation, SVM model training, and prediction are recorded and shown in Fig. 13. Especially, as shown in the stacked bars, the runtime of MLFP increases significantly with the number of mixture components of ΓMM. Compared with the other parts, fitting a suitable ΓMM requires a very large amount of runtime. Fortunately, a better ΓMM can be fit effectively in an offline manner with sufficient intensity values. Despite this, a small number of mixture components should be sufficient for some online applications when striking a balance between the average *o.a.* and runtime.

Third, the classification performance of the proposed MLFP is also checked with respect to different settings of local scale values scl. Like the former experiment, 50% of the superpixels selected randomly from the Rosenheim image are also utilized for SVM model training, while the remaining 50% are utilized for model testing. In this experiment, eight different settings of scl are checked in this experiment, which are increased by adding one scale value progressively from one to 15 in two-value intervals. In addition, the number of

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TABLE IV

AVERAGE *o.a.* AND κ OF PROPOSED MLFP WITH RESPECT TO DIFFERENT NUMBERS OF SUBCOMPONENTS IN ΓMM

K	**2**	**4**	**8**	**12**	**16**	**20**
o.a. (%)	28.05 ± 0.71	80.09 ± 0.59	88.14 ± 0.48	**88.71** ± 0.54	88.63 ± 0.52	88.37 ± 0.46
κ	0	0.7562 ± 0.0031	0.8716 ± 0.0051	**0.8809** ± 0.0045	0.8801 ± 0.0048	0.8768 ± 0.0052
K	**24**	**28**	**32**	**36**	**40**	**60**
o.a. (%)	88.62 ± 0.63	88.29 ± 0.55	88.56 ± 0.49	88.04 ± 0.39	88.24 ± 0.63	88.21 ± 0.44
κ	0.8800 ± 0.0059	0.8769 ± 0.0045	0.8473 ± 0.0047	0.8392 ± 0.0056	0.8432 ± 0.0050	0.8435 ± 0.0045

TABLE V

AVERAGE OVERALL ACCURACIES (*o.a.*), COHEN'S KAPPA COEFFICIENTS (κ), AND RUNTIME FOR PROPOSED MLFP WITH RESPECT TO DIFFERENT SETTINGS OF *scl*

scl	***o.a.***(%)	κ	**Runtime**(s)		
			ΓMM fitting	**FV computation**	**MLFP generation**
[1]	87.64 ± 0.03	0.8659 ± 0.0040	361.70 ± 15.83	11.50 ± 0.35	2.33 ± 0.03
[1, 3]	88.36 ± 0.07	0.8743 ± 0.0092			4.23 ± 0.03
[1, 3, 5]	88.54 ± 0.01	0.8766 ± 0.0061			6.13 ± 0.05
[1, 3, 5, 7]	88.85 ± 0.07	0.8774 ± 0.0073			8.07 ± 0.09
[1, 3, 5, 7, 9]	89.35 ± 0.09	0.8851 ± 0.0089			10.09 ± 0.07
[1, 3, 5, 7, 9, 11]	**89.52** ± 0.11	**0.8853** ± 0.0066			11.23 ± 0.13
[1, 3, 5, 7, 9, 11, 13]	89.13 ± 0.17	0.8792 ± 0.0087			14.31 ± 0.09
[1, 3, 5, 7, 9, 11, 13, 15]	88.93 ± 0.15	0.8781 ± 0.0046			17.57 ± 0.19

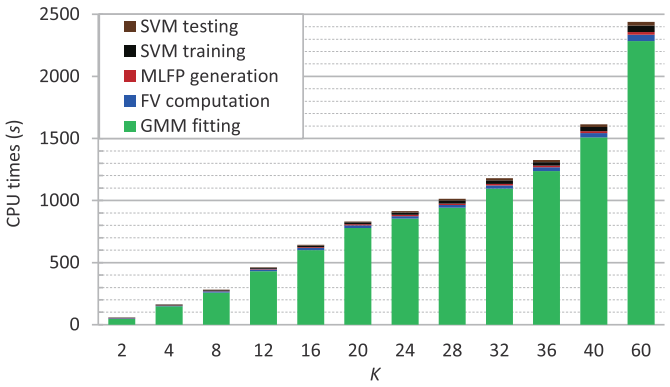


Fig. 13. Runtimes of different stages in land-cover classification of proposed MLFP with respect to number of subcomponents in ΓMM.

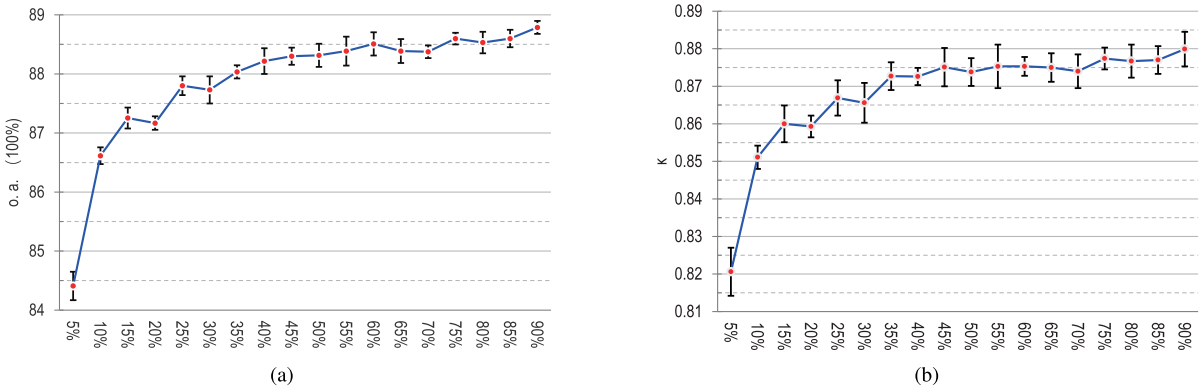
subcomponents in ΓMM is set as 12. Similar to the former experiment, the average overall accuracy (*o.a*), Cohen's kappa coefficient (κ), and runtime are recorded and listed in Table V after this experiment was conducted ten times independently. It can be observed that the average *o.a.* increases with the number of scale values, where the highest *o.a.* value is reached when scl $= [1, 3, 5, 7, 9, 11]$. A larger scale value for MLFP computation can capture rich spatial structure information in addition to the statistical characteristics for different land covers. Compared with a small scale size, it can be observed that the runtime for the MLFP does not increase excessively when the scale increases.

2) Sensitivity to Number of Labeled Samples: As a supervised approach, the sensitivity of the proposed MLFP feature learning approach is also checked in this experiment. After this experiment was conducted ten times independently, the average overall accuracies (*o.a.*) and Cohen's kappa coefficients (κ) are checked with different numbers of labeled superpixels i.e., from 5% to 90%. The parameters of the proposed approach in this experiment are set as $K = 12$ and scl $= [1, 3, 5]$, respectively. It can be observed that the overall accuracy values [shown in Fig. 14(a)] of the proposed approach increase with the number of training superpixels. Similar results can be observed from Cohen's kappa coefficients, as shown in Fig. 14(b). In this experiment, the highest *o.a.* of the proposed approach is reached with 90% training superpixels. In addition, it can be observed that *o.a.* is higher than 88% when the training number of superpixels is higher than 35%. Even with 5% training superpixels, *o.a.* of the proposed MLFP is higher than 80%, which indicates that the proposed MLFP features can effectively capture enough discriminant information of different land covers.

3) Sensitivity to Classifier Selection: In the last experiment, the other three classic classifiers, K-Nearest Neighbor (KNN), softmax, and random forest, are also employed to evaluate the discriminant capability when characterizing different land covers of the proposed approach. The detailed setting information of these classifiers is listed as following.

1) For KNN, a five-nearest neighbors' classifier using Euclidean distance is trained as a benchmark approach for classification performance comparison.
2) For softmax, a single-layer neural network with a softmax layer is trained to label different land covers. The loss function for the softmax layer is setting as cross entropy, and the maximum number of training iterations is setting as 1000.
3) For random forest, ten decision trees with minimum leaf size 1 are bagged together to construct a random forest classifier for labeling different land covers.

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(a) (b)

Fig. 14. Overall accuracies and Cohen's kappa coefficients of MLFP with different training numbers of superpixels. (a) *o.a.*. b) κ.

TABLE VI

AVERAGE OVERALL ACCURACIES (*o.a.*) AND COHEN'S KAPPA COEFFICIENTS (κ) FOR PROPOSED MLFP WITH RESPECT TO DIFFERENT CLASSIFIERS

	features	patch vector	textures	Gabor filters	BoW	SAE	MLFP
o.a.(%)	***KNN***	69.38 ± 0.73	58.07 ± 0.59	81.17 ± 0.61	82.21 ± 0.65	85.32 ± 0.36	**85.60** ± 0.51
	random forest	87.05 ± 0.34	83.97 ± 0.43	87.43 ± 0.39	86.26 ± 0.41	91.35 ± 0.25	**93.74** ± 0.47
	softmax	39.64 ± 0.97	49.33 ± 0.63	77.78 ± 0.87	78.27 ± 0.79	85.84 ± 0.72	**86.10** ± 0.61
	SVM	65.55 ± 0.46	57.76 ± 0.45	78.21 ± 0.25	82.20 ± 0.08	83.22 ± 0.21	**88.33** ± 0.06
κ	***KNN***	0.6012 ± 0.0037	0.5132 ± 0.0029	0.7783 ± 0.0035	0.7857 ± 0.0042	**0.8253** ± 0.0043	0.8079 ± 0.0037
	random forest	0.8677 ± 0.0031	0.8524 ± 0.0043	0.8601 ± 0.0038	0.8407 ± 0.0033	0.9211 ± 0.0045	**0.9390** ± 0.0038
	softmax	0.1683 ± 0.0037	0.2777 ± 0.0044	0.7203 ± 0.0033	0.7256 ± 0.0038	0.8507 ± 0.0041	**0.8555** ± 0.0042
	SVM	0.5182 ± 0.0072	0.3336 ± 0.0050	0.7049 ± 0.0025	0.7825 ± 0.0019	0.8057 ± 0.0030	**0.8746** ± 0.0013

For performance comparison, 50% superpixels randomly selected from the Rosenheim image are utilized for classifier training, while the remaining 50% superpixels for testing. After conducting this experiment with the same settings ten times independently, the average overall accuracy (*o.a.*) and Cohen's kappa coefficient (κ) are listed in Table VI, respectively. For the convenience of comparison, the results of SVM in Table II with different feature vectors are also relisted in this table.

It can be observed that the proposed MLFP yields the highest *o.a.* values over the four selected classifiers. At the same time, the random forest classifier reaches the best classification accuracies and Cohen's kappa value among these four selected classifiers. This is caused by the fact that the random forest classifier is an ensemble of several decision trees, which can fully exploit the discriminant information from each decision tree while labeling different land covers. In addition to the softmax classifier, the KNN, random forest, and SVM perform well on all compared features, where the average overall accuracy values of these classifiers are larger than 50%. Furthermore, for features generated from the Gabor filters, BoW, SAE, and the proposed MLFP, both the average overall accuracies and Cohen's kappa coefficients are relatively high. It indicates that discriminant information of different land covers could be effectively captured by these comparatively complicated feature vectors. Since both the information of the statistical distribution of SAR image intensities and the information of the spatial structure are exploited, the proposed MLFP obtains the best performance when distinguishing different land covers.

IV. CONCLUSION

In this article, an MLFP is proposed to characterize the content of high-resolution SAR images. After the modeling of the probability density of high-resolution SAR intensities via a Gamma mixture model, the generative and discriminant information can be exploited effectively via FVs. Meanwhile, the multiscale spatial structure information within the specified neighborhood can also be exploited to characterize the concerned pixels. As two experiments on land-cover classification and scene category have shown, the proposed approach has excellent capacity in the characterization of different land covers. In addition, the computation burden of MLFP extraction does not increase substantially compared with that of the other approaches.

However, there are still some issues of the proposed approach that should be investigated further. Especially, they are listed as follows.

1) Due to the influence of LFP computation and superpixels segmentation, the proposed MLFP does not have shift and rotation invariance.
2) The number of subcomponents of ΓMM fitting in this work is determined artificially according to the experimental results, which seriously limits the flexibility of the proposed method.

In order to overcome these limitations, an appropriate selection strategy of subcomponent number in ΓMM should be studied in future works. Also, a general model will be provided from SAR images acquired by different platforms in various imaging modes. Moreover, some adaptive strategies of parameter

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selection, such as adaptive neighbor weights learning for LFP computation, should be considered in future works. In addition, as the basic feature vector of different land covers, the MLFP should be studied further after its incorporation with some modern high-level and abstract feature learning techniques.

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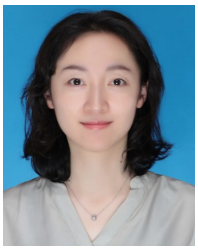
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